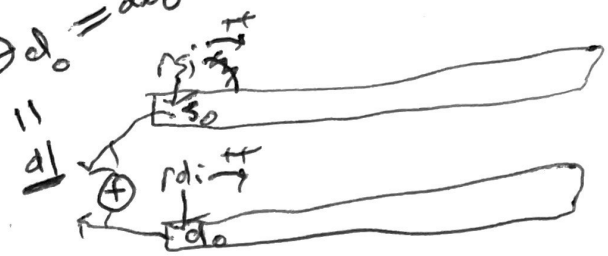
$s_0 \oplus d_0 = dl_0$



$al_1 = al_0 \vee dl_0 = al_0 \vee (s_0 \oplus d_0)$

$al_2 = al_1 \vee (s_1 \oplus d_1) = (al_0 \vee (s_0 \oplus d_0)) \vee (s_1 \oplus d_1)$

$al_n = al_0 \vee \bigvee_{j=0}^{rcx-1} (s_j \oplus d_j) = 0 \Leftrightarrow [rsi+i] = [rdi+i] \;\; \forall i \in [0, rcx]$

difference check

```
(rcx) .loop:
        mov     dl,byte [rsi]
        xor     dl,byte [rdi]
        inc     rsi
        inc     rdi
        or      al,dl
        loop    .loop
```

(suggested by math PhD friend Kirsh, pedantry)

Δ

$\|$

0

$y_{[rsi+i]} : C^{\vee} \to Set$ (cat theory + group theory)

$y_{[rsi+i]}([rdi+i]) = Hom_C([rsi+i], [rdi+i])$

$y^{[rsi+i]}([rdi+i]) = Hom_C([rdi+i], [rsi+i])$

(group $G \to Sym(G)$, $\lambda_g(h) = g \cdot h$ } group homomorphism)

recall: (category (obj, mor(hom), composition, id))

"wiley"

(explicit functor)

(up to unique isomorphism)

Yoneda lemma

obj $\in$ cat is completely determined, by how it looks from every possible test object (collection of all morphisms into or out of it)

ie $\forall$ obj $A \in C$, define contravariant representable functor $y_A : C^{op} \to Set$

ie Yoneda embedding:

$y_A(X) = Hom_C(X, A)$

$y^A(X) = Hom_C(A, X)$

(i) $\forall$ functor $F : C^{op} \to Set$,

$Nat(y_A, F) \cong F(A)$ "objects"

(ii) $y : C \to [C^{op}, Set]$ is fully faithful

ie $\forall$ obj A $(A, B \cong Nat(y_A, y_B))$